Week 6

Odometry, Fusion, and Teams

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The individual block ends today

Five labs done. From here: **four teams, one vehicle.**

Today	Teams announced · charter workshop
10/26	Charters due
11/02	Mid-semester presentation
11/23 – 12/14	The merge ladder

1. Odometry

Integrating your way to a position

$$\dot{x} = v\cos\theta \qquad \dot{y} = v\sin\theta \qquad \dot{\theta} = \frac{v\tan\delta}{L}$$

Two constants turn raw counts into those quantities:

- `meters_per_tick`
- steering counts → radians, from the absolute sensor

MRider refuses to derive `meters_per_tick`

Roll-out calibration:

- Drive a **measured** straight distance **L**
- Record the tick delta
- `meters_per_tick = L / Δticks`
- Three times, averaged

Authoritative because it bypasses guessing.

No gear ratio. No quadrature decode factor. No tyre diameter under load.
All of them fold into one measured number.

Which is why F7 can forbid a PPR figure

The source project:

- firmware pins **52 PPR**
- its own BOM lists a **16 PPR** motor

A factor-of-three disagreement nobody caught.

Measure yours. The roll-out makes the resulting distance safe either way — that is exactly why the calibration is designed that way.

Why it drifts

Error	Effect
Wheel slip	Rotation without ground motion — always over-estimates
Tyre deformation	6 kg of payload on a 10 kg vehicle
Gearbox backlash	The encoder is upstream and cannot see it
Steering angle error	**Heading error — grows unbounded**

1° of heading bias = 0.35 m off after 20 m.

And it keeps growing whether or not you keep making the error.

MRider's measurement is narrower still

- Both rear motors **in parallel** on one driver channel
- Only **one** carries an encoder — ADR C
- Measurement taken at **one motor shaft**

In a turn, the two rear wheels genuinely travel different distances.

One of them is simply not observed. Its speed is inferred.

2. The EKF

Fuse velocities, not pose

```
odom0_config: [false, false, false,   # x, y, z   <- NOT fused
               false, false, false,   # roll, pitch, yaw
               true,  true,  false,   # vx, vy, vz  <- fused
               false, false, true,    # vyaw
               false, false, false]
```

> **Integrated wheel position drifts without bound and there is no absolute reference indoors to correct it.**

Feed it an already-drifted pose and it fuses the drift **as evidence.**

Take the IMU's yaw, not its accelerations

> **On a vibrating ride-on chassis they are mostly drivetrain buzz.**

A sensor axis you **can** read is not automatically one you **should** fuse.

Also `imu0_relative: true` — no magnetometer heading reference is trusted indoors, near steel and motors.

The Lisbon declination

B-MROVER's `navsat_transform` carried a magnetic declination set for **Lisbon**.

Wrong here — and it would **silently rotate the global frame.**

MRider's indoor profile does not fix the number.
It drops the GNSS EKF entirely.

The choice was not between a right number and a wrong number. It was between an unverifiable number and deleting the thing that needed it.

Remember this move. It comes back in Week 8.

Transform ownership

Transform	Owner
`map → odom`	`slam_toolbox`
`odom → base_link`	EKF
`base_link → *`	`robot_state_publisher`

`enable_odom_tf: false` on the controller, so the EKF is the **sole** publisher.

The gates this serves:

≤ 2% drift over 20 m straight · ≤ 5° heading error on a closed 20 m figure-eight

Nobody has measured either. That is `mitt_bench` — Simulation & Validation.

3. Teams

The four tracks

Track	Ships something that
E — Electrical & Firmware	`firmware/` ; `DbwCommand` / `DbwStatus` actually flowing
C — Chassis & Measurement	A **measured** `mitt_dimensions.yaml` ; M1/M2/M3
S — Simulation & Validation	`mitt_bench` ; tests that are not linters; CI
W — Software & Autonomy	`mitt_hardware` ; odometry node; perception

Charter — due 10/26

1. **Who owns what** — a **person's name** against every deliverable.
   Not "the team will"

2. **Definition of done** — the observable evidence
   "The encoder is mounted" is not evidence
   "Monotonic counts through a full sweep, logged, plotted, committed" is

3. **What you need from other tracks, and by when** — with dates

4. **Your risks** — two or three, and what you would do instead

Write it in the room, today

The other three tracks are here. **Three of the four sections are about them.**

- Software needs measured geometry from Chassis
- Simulation needs it too
- Electrical needs bench kits and column dimensions

If those asks are not dated today, they get discovered in Week 13.

And Week 13 is too late.

Standups start today

Last 30 minutes, every session. Each team, briefly:

- What moved
- What is blocked
- **What you need from whom**

This is where the participation grade is observed — and it is assessed on **honest reporting**, including reporting that nothing moved.

Next week

Embedded ROS 2 — micro-ROS, the Teensy, and `hardware_interface`.

The Electrical track needs it immediately; everyone gets it, because that boundary is where every other track's work eventually lands.

Reading: M2 and dbw.md §10–12.

Charters due 10/26.